Dexin Ren

M.S. Computer Science | Data-Centric & Reliable ML for Geospatial and Health Applications

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Research Focus

Data-centric and reliable machine learning: robustness under distribution shift, model failure analysis, and generalization in geospatial and health applications. I study how data characteristics, learning objectives, and evaluation design shape what a model actually learns.

Education

University of Southern California Jan 2025 – Dec 2026
M.S. Computer Science (General), GPA: 3.7/4.0
Research advisor: Prof. Cyrus Shahabi (USC InfoLab)

University of Illinois Urbana-Champaign Aug 2021 – May 2024
B.S. Mathematics + Computer Science, GPA: 3.62/4.0, High Distinction

Publications

Dexin Ren, Shang-Ling Hsu, Chen Chu, Lifeng Lin, Ryosuke Shibasaki, Noboru Koshizuka, and Cyrus Shahabi. *RectifiedTraj: Map-Free Causal GPS Trajectory Denoising with Rectified Flow.* **ACM SIGSPATIAL '26**, short paper.
DOI | Preprint

Dexin Ren, Shang-Ling Hsu, Chen Chu, Lifeng Lin, Ryosuke Shibasaki, Noboru Koshizuka, and Cyrus Shahabi. *RectifiedTraj Revisited: An Expanded Evaluation of Map-Free Causal GPS Trajectory Denoising.* **UMFM '26** (ACM SIGSPATIAL workshop), full paper. **Best UMFM Workshop Paper Award**.
DOI | Preprint

Hongyuan Dong, Jiaqing Xie, Zhi Jing, and **Dexin Ren**.
Variational Autoencoder for Anti-Cancer Drug Response Prediction. AI4PH Workshop at ICLR 2021.
arXiv:2008.09763

Research Experience

RectifiedTraj: Causal GPS Trajectory Denoising Aug 2025 – Sep 2026
Researcher / First-Author Lead, USC InfoLab

- Designed RectifiedTraj, a map-free, causal GPS denoiser that predicts a per-point correction field with one-step rectified flow, built for streaming use on mobile devices (**124 ms** per 256-point chunk on a 2020 Pixel 4a, CPU only).
- Traced through failure analysis of prediction-error distributions why direct-coordinate models break as trajectories cover more distance, and showed that predicting the correction keeps training and inference stable across datasets.
- Designed a head-buckle windowing scheme for causal streaming that reduced error by **21%** at a 24% throughput cost (up to 30% at the extreme setting).
- Designed robustness evaluations across datasets, geography, simulators, noise conditions, and sampling intervals to test generalization beyond training data.
- Demonstrated strong zero-shot generalization across unseen datasets, noise distributions, and GPS sampling intervals, improving accuracy on real-world data by **12.42 percentage points** (**44.63% to 57.05%**).

NeurogenesisGNN: EEG-Based Neurogenesis Estimation Jun 2026 – Present

Part-Time Research Assistant, USC IMSC & Bonaguidi Lab

- Built the end-to-end pipeline estimating patient-level hippocampal neurogenesis from EEG and clinical variables, using an existing EEG graph model as a frozen feature extractor.
- Treated ∼**13,000 clips from only 30 patients** (**6,400+ features per clip**) as a small-sample, high-dimensional problem, using PLS + Ridge and a learned percentile calibrator per patient.
- Detected patient-identity leakage in clip-level validation: with patient-split cross-validation and retraining for every ablation, the apparent importance of disease duration vanished (+12.3 to +0.9 pp), indicating the earlier model was memorizing patients.
- Ran controlled patient-level experiments on encoders, single channels, and channel groups; channels on the tissue-sampling side carried the most signal, and mixing seizure with non-seizure clips lowered error from ∼68–72% to ∼58%.
- Analyzed the zero-neurogenesis subgroup with hurdle models; current features cannot yet identify it.

HAYSTAC: Human Mobility Anomaly Detection Jun 2026 – Sep 2026

Research Team Member, USC InfoLab & Novateur Research Solutions

- Ran controlled experiments across architectures, context-window sizes, network depths, and loss functions to diagnose why a specialized Transformer underperformed a simpler MLP baseline.
- Investigated class-imbalance failure modes in sparse-anomaly detection, where near-zero anomaly scores yield low training loss without localizing anomalies.
- Showed that synthetic-anomaly training loss is a poor guide for checkpoint selection (asymmetric focal loss agreed best with detection performance), and that validating on the same synthetic labels hides overtraining.
- Redesigned the training data pipeline, roughly halving memory use and speeding up training by **2**–**12**×.

TrafficWhatIf / DeepCitySim: Intervention-Aware Traffic Prediction Sep 2026 – Present

Team Lead / Researcher, USC InfoLab

- Developing a modular MATSim- and SUMO-based data generation pipeline that produces matched what-if scenarios across traffic policies, agent distributions, and cities.
- Modeling agents with prior travel experience to study how established mobility patterns respond to unexpected traffic-policy interventions.

Technical Skills

Programming: Python, C/C++

Other tools: Chosen per project as the research requires; quick to learn new frameworks and methods.